SUCCESS IS NOT IDENTIFICATION: CROSSED WORLD INTERVENTIONS FOR ATTRIBUTING AGENT POLICIES

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ABSTRACT

Task success does not identify why an interactive policy succeeds: the same action can follow an operative mechanism, a correlated visible cue, or a fixed prior. We introduce a crossed-intervention assay that independently moves a hidden tool affordance and a visible color cue across matched worlds. Agents act through query and commit macro-actions that compile to legal environment trajectories, allowing behavioral dependence to be measured without requiring verbal explanations. The assay reports mechanism and cue tracking, paired responsiveness, cue susceptibility, evidence acquisition and use, natural commitment coverage, and task success as an unaggregated vector. Exact scripted controls, balanced tool roles, paired seed-level inference, and lossless provenance make construct failures observable. Two behavior-cloned policies are indistinguishable by success on cue-aligned worlds but occupy opposite corners of the attribution plane on 64 held-out crossed worlds, replicated across five initializations per policy class. In uncached developmental pilots, two language models passed the deterministic serving gate: neutral prompts placed both near the mechanism-following corner, whereas either of two independently seeded cue-demonstration prefixes moved both to the cue-following corner. These pilots triggered a prospectively specified 3×2 challenge and fixed the protocol; confirmatory outcomes remain unobserved. The claims concern behavioral policy dependence, not internal representations or causal knowledge.

1 INTRODUCTION

A successful terminal action is compatible with several policies. An agent may have used evidence about the mechanism, followed a visible feature that happened to correlate with success, or repeated a preferred action. Observational task success cannot distinguish these explanations. This under-identification is especially acute in interactive benchmarks, where discovery, navigation, action grounding, and terminal choice are often bundled into one outcome.

We study an intentionally narrower construct: which controlled world variable a policy's behavior tracks. For every base world, we independently move the operative conductivity bit and a red color cue between two focal tools while holding public nuisance variables fixed. This 2×2 design turns policy attribution into a paired behavioral measurement. A conductivity detector supplies public evidence; the assay never asks the agent to describe a belief or reveal a chain of thought.

The primary interface contains only QUERY and COMMIT macro-actions. Each is compiled into legal movement, pickup, and application actions and executed by the underlying environment. This separates epistemic choice from low-level execution while retaining an auditable environment trace. We make no claim about internal causal representations. Our contribution is a controlled instrument for measuring behavioral policy dependence, together with integrity tests, reference policies, and paired inference.

An earlier public prototype bundled interventions with availability-dependent diagnostics. The present protocol replaces both with crossed interventions and availability-invariant estimands. The earlier prototype is treated as related public work and corrected explicitly without using it as v0.2 evidence.

Proposition 1 (aligned success is non-identifying). Let $M, C \in \{0, 1\}$ denote the mechanism and cue carriers, and consider the mechanism policy $A_M(M, C) = M$ and cue policy $A_C(M, C) = C$.

If evaluation is restricted to aligned worlds, $M = C$, then $A_M(M, C) = A_C(M, C)$ and the policies have identical task success. Task success on the aligned support therefore cannot identify which variable controls action. On crossed worlds, $M \neq C$ implies $A_M(M, C) \neq A_C(M, C)$, so the policies become behaviorally distinguishable. The same argument applies to any two policies that agree on observational support but differ under a controlled intervention. □

2 Related work

Shortcut learning shows that predictive success can arise from unintended surface regularities rather than the target mechanism Geirhos et al. (2020). Work on invariance and distribution shift seeks predictors that remain stable across environments Arjovsky et al. (2019); Koh et al. (2021); our concern is narrower and interactive, asking which independently manipulated variable controls a policy's terminal action within paired worlds.

Causal representation and causal-reasoning research studies when learned systems recover or exploit causal structure Pearl (2009); Schölkopf et al. (2021); Dasgupta et al. (2019). The present assay does not infer a representation. It measures observable acquisition and use of mechanism-specific evidence. This claim boundary also distinguishes the assay from broad language-model and agent benchmark suites Liang et al. (2023); Liu et al. (2024), where aggregate success can bundle discovery, grounding, execution, and attribution.

Interactive environments such as TextWorld, BabyAI, and ALFWorld expose grounded sequential behavior Côté et al. (2018); Chevalier-Boisvert et al. (2019); Shridhar et al. (2021). Our macro-actions intentionally factor low-level execution out of the primary estimand while retaining a compiled legal trace, making the crossed epistemic choice the unit of analysis.

Several contemporary benchmarks share the premise that success need not identify mechanism recovery. CausaLab compares predictive success with fidelity of a recovered causal graph and structural equations Yang et al. (2026). ZendoWorld evaluates active visual concept induction and finds that label accuracy does not imply recovery of the hidden logical rule Koehler et al. (2026). CausalGame asks agents to design experiments and produce explanation reports under selection bias, measurement error, and hidden confounding Chen et al. (2026). These benchmarks score an externalized graph, equation set, logical rule, or causal explanation. Our assay instead attributes dependence from changes in action under crossed interventions, so it applies to policies that cannot or do not externalize a hypothesis.

The GUI-agent state-reliance assay is methodologically closer: it uses paired interventions and deterministic forced choice to attribute beliefs to pixels, serialized structure, or priors Zhang et al. (2026). It manipulates redundant observation channels for a state belief. Our setting instead crosses a hidden operative mechanism with a correlated visible cue in a sequential query–commit environment and retains abstention as a measured outcome.

3 Crossed behavioral assay

Worlds. Each base seed defines three stable tool objects, a detector, and a gate. A seed-cycled permutation selects two focal global slots and one neutral slot, so no global slot is permanently excluded. The local mechanism condition $m \in \{0, 1\}$ selects which focal tool is conductive; the cue condition $c \in \{0, 1\}$ selects which focal tool is red. The other focal tool is blue and the neutral tool is green. Positions, shapes, textures, sizes, object and candidate order, and anonymized labels remain fixed across the four cells.

The detector returns `signal-positive` exactly when the queried tool is conductive. All other latent tool properties are false. Detector applications to non-tools and repeated queries are absent from the primary action menu.

Macro-actions. At each decision the policy chooses $\textsc{Query}(i)$ or $\textsc{Commit}(i)$ for tool slot i. A deterministic shortest-path compiler emits only legal movement, pickup, detector-application, and gate-application actions, all executed through the environment transition function. The first commit terminates the episode whether it succeeds or fails. After three distinct queries only commit actions

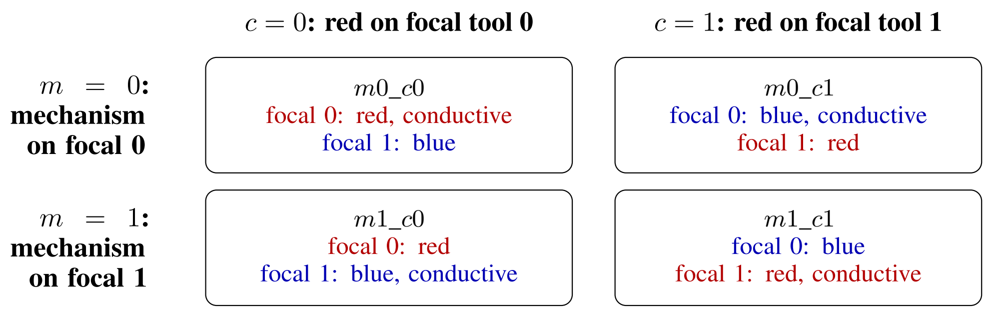

Figure 1: The crossed quartet. Mechanism rows change only hidden conductivity; cue columns swap red and blue. The green neutral tool and every other public feature are fixed within a base world. "Conductive" is shown here for exposition and is never included in the agent's observation.

remain. A response that cannot be mapped to an available action terminates with $A = \perp$. Parse-failure rate and natural commitment coverage are reported explicitly; the evaluator never imputes a tool.

Estimands. Let $A_{m,c} \in \{0, 1, 2, \perp\}$ be terminal choice; let $g_{m,c}$ and $r_{m,c}$ denote the global mechanism and red-cue carriers; and define $D_{m,c} = \mathbf{1}[A_{m,c} \neq \perp]$. Tracking is conditional on commitment:

$$T_M = \frac{\sum_{m,c} D_{m,c}\mathbf{1}[A_{m,c} = g_{m,c}]}{\sum_{m,c} D_{m,c}}, \qquad T_C = \frac{\sum_{m,c} D_{m,c}\mathbf{1}[A_{m,c} = r_{m,c}]}{\sum_{m,c} D_{m,c}}.$$

Strict mechanism responsiveness is also commitment-conditional:

$$R_M = \frac{\sum_c D_{0,c}D_{1,c}\mathbf{1}[A_{0,c} = g_{0,c} \wedge A_{1,c} = g_{1,c}]}{\sum_c D_{0,c}D_{1,c}}.$$

We retain $\perp$ as an ordinary outcome for unconditional cue susceptibility,

$$S_C^{\text{all}} = \frac{1}{2}\sum_m \mathbf{1}[A_{m,0} \neq A_{m,1}],$$

and separately report the commitment-conditional form

$$S_C^{\text{commit}} = \frac{\sum_m D_{m,0}D_{m,1}\mathbf{1}[A_{m,0} \neq A_{m,1}]}{\sum_m D_{m,0}D_{m,1}}.$$

Every conditional denominator is emitted. Coverage is $\frac{1}{4}\sum_{m,c} D_{m,c}$, while unconditional choice accuracy is $\frac{1}{4}\sum_{m,c}\mathbf{1}[A_{m,c} = g_{m,c}]$, so abstention is counted as incorrect only in the latter. We additionally report natural commitment coverage, aligned, crossed, quartet, and cell success, detector-query rate, unique queries, identification coverage, evidence-consistent commitment, and first-query cue bias. These quantities are never collapsed into one score.

The paired base seed is the statistical unit. Intervals use 10,000 seed-clustered bootstrap resamples and every conditional estimate reports its denominator. The two analytic references, neither uniquely called "chance": uniform commitment over all three tools gives $(R_M, S_C, T_C) = (1/9, 2/3, 1/3)$, while uniform commitment over the two focal tools gives $(1/4, 1/2, 1/2)$.

Promoted 3×2 challenge. The 2×2 quartet is the construct-validation assay. A prospectively specified pilot rule promoted a harder language-model confirmatory variant after neutral-arm mechanism tracking and query rate both exceeded their ceiling-risk thresholds. In this variant, $m \in \{0, 1, 2\}$ allows conductivity to occupy any tool, including the otherwise green tool, while $c \in \{0, 1\}$ still moves red between the two focal tools. The six cells share all nuisance variables. Primary outcomes are commitment-conditional mechanism accuracy and cue following, commitment-conditional cue susceptibility across the three cue pairs, detector query rate, coverage, all-six-cell success, and cell success. A complete six-cell base seed remains the bootstrap unit; 10,000 seed-clustered resamples and explicit conditional denominators are used throughout. The BC construct validation remains on the original quartet and is not pooled with the language-model challenge.

Validation	Scale	Result
Quartet invariants	10,000 base seeds	0 violations
Legal oracle execution	8,000 cell–policy runs	100% success
Mechanism-axis public equality	all checked pairs	100%
Cue-axis equality after color mask	all checked pairs	100%
Scripted policy calibration	all reference policies	exact coordinates

Table 1: Pre-scientific integrity gates. Scientific outcomes cannot waive a failed row.

4 Integrity controls

The mechanism-axis prompts are byte-identical, including color. The cue-axis public states are identical after masking only the manipulated color field. Stable IDs never encode condition. Candidate order is canonical. The neutral global slot, mechanism carrier, and cue carrier are balanced across seeds.

Reference policies include a hidden-mechanism oracle, a public detector-then-commit policy, cue and anti-cue followers, a fixed global-slot policy, an abstainer, a uniform three-tool committer, and a focal-uniform committer. The random policies are plotted at their analytic nonzero reference coordinates. A legal certificate suite executes through the environment; no oracle relocates objects or bypasses navigation.

The released audit exercised 10,000 base seeds and 1,000 oracle seeds. All 8,000 cell–certificate executions used legal environment actions, remained within budget, and succeeded. No invariant or control-coordinate violation occurred (Table 1).

Trace schema 0.2 records condition axes, stable world and base-spec hashes, repository and protocol manifests, model and tokenizer revisions, launch settings, prompt/candidate/label hashes, requested and effective generation parameters, usage, finish and provider status, parse status, visible response, detector evidence, terminal choice, and the lossless low-level ledger. A scientific result cannot waive a failed integrity check.

5 Evaluation design

Pilot and confirmatory manifests use disjoint seeds. Prompt engineering, parser changes, serving preflight, and per-model output-ceiling selection occur only on the pilot. The selected ceiling is the smallest value for which fewer than one percent of pilot completions terminate for length and action parse rate has plateaued. Serving validation separately requires zero response, parsed-action, and candidate-set conflicts among independent calls with identical full visible policy inputs. Failed runs emit no scientific summary. Any integrity repair after freezing requires a complete rerun of every affected confirmatory condition.

Because visible assistant responses are retained in subsequent full-history prompts, response-level nondeterminism can alter later decisions even when the current parsed action agrees. The full-history protocol therefore requires identical responses for identical full policy inputs.

We use disjoint pilot and confirmatory designs. Before confirmatory evaluation, we prospectively froze the models, seeds, prompts, output ceilings, serving manifests, estimands, exclusion rules, and analysis procedure against the clean tagged protocol candidate. Confirmatory outcomes had not been acquired when this draft was built. The panel contains Qwen3.6-27B and Mistral-Small-3.2-24B, the two models whose pilot serving paths passed the gate; GPT-OSS-20B is excluded by the declared serving-integrity rule. The confirmatory panel uses the same 64 base seeds, 100–163, for the neutral arm and for each of two fixed cue and two fixed mechanism demonstration pools. Cue and mechanism prefixes contain six examples each. Each mechanism prefix covers a complete 3×2 block, making red, blue, and green positive exactly twice; cue prefixes contain six independently generated aligned red-success examples. The pools are fixed replications rather than a population sample of arbitrary prompts. The smallest substantively meaningful change in cue or mechanism tracking is 0.15; a null is informative only when its interval excludes that magnitude.

Policy	Align.	Crossed	R_M	S_C^{commit}	Tracking	Query rate
Cue-BC	1.00	0.00	0.00	1.00	$T_C = 1.00$	0.00
Mechanism-BC	1.00	1.00	1.00	0.00	$T_M = 1.00$	1.00

Table 2: Held-out construct validation on base seeds 100–163. Every value was reproduced across all five initialization seeds in each regime.

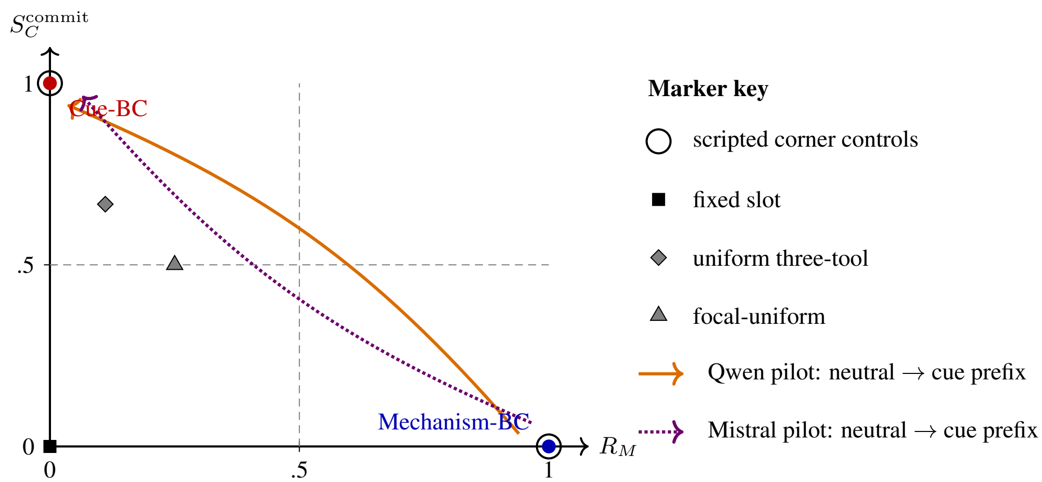

Figure 2: Construct validation and developmental 2×2 model pilots. Cue-BC and Mechanism-BC reproduce opposite corners on every held-out initialization. Both model pilots overlap the mechanism corner under neutral prompts and the cue corner under either cue-demonstration prefix; mechanism prefixes remain at or near the mechanism corner. Cue and anti-cue controls share (R_M, S_C) and are separated by T_C, so the full metric vector remains necessary.

Behavior-cloning controls use the same architecture, initialization-seed set, optimizer, learning-rate schedule, number of supervised state–action examples, updates, and evaluation worlds. Cue-BC learns direct cue-correlated commits; Mechanism-BC learns queries followed by evidence-consistent commits. The regimes also differ in evidence and action distributions, so this comparison is not described as changing demonstration policy alone. Five initialization seeds and 4,096 supervised state–action transitions per model run are used.

6 Results

6.1 Construct validation with learned non-verbal policies

We trained five independently initialized behavior-cloned policies under each of two regimes, using the same architecture, optimizer, number of supervised state–action transitions, updates, and evaluation worlds. Both policy classes achieved 1.00 success when cue and mechanism were aligned, so aligned task success did not distinguish them. Their behavior separated completely when the factors were crossed (Table 2).

Cue-BC committed directly to the visible red carrier: $R_M = 0$, $S_C^{\text{commit}} = 1$, cue tracking 1.00, and detector-query rate 0. Mechanism-BC queried before committing and solved aligned and crossed cells: $R_M = 1$, $S_C^{\text{commit}} = 0$, mechanism tracking 1.00, and detector-query rate 1.00. Figure 2 shows that the two learned policies occupy opposite reference-policy corners.

The regimes differ in evidence availability and action distribution. This result validates recovery of known behavioral policy classes; it does not identify the isolated causal effect of demonstration semantics and does not establish internal causal knowledge.

6.2 Language-model serving pilots and frozen confirmatory study

Qwen3.6-27B and Mistral-Small-3.2-24B each passed an uncached, serial serving pilot: every replicated full-policy input produced an identical visible response, parsed action, and candidate set. In the 16-seed 2×2 pilot, both neutral arms had full coverage, mechanism tracking 1.00, cue susceptibility 0, quartet success 1.00, and detector-query rate 1.00. Either independently seeded cue prefix moved both models to mechanism tracking 0.50, cue susceptibility 1.00, quartet success 0, and query rate 0. Mistral's two mechanism-prefix replications remained at or within 0.016 of the mechanism corner. Qwen's mechanism-prefix choices were mechanism-responsive whenever it committed, but 12 length terminations in one replication selected a 2,048-token ceiling using the pilot-only rule. These are developmental manipulation checks, not confirmatory effect estimates.

Both neutral arms crossed the prospectively specified ceiling-risk threshold, so we ran the disjoint 12-seed 3×2 challenge pilot and selected that variant before freezing. All 144 model–world cells produced commitments. Mistral achieved mechanism accuracy 1.00 (95% CI $[1.00, 1.00]$), cue susceptibility 0, query rate 1.00, and all-six-cell success 1.00. Qwen achieved mechanism accuracy 0.944 ($[0.861, 1.00]$), cue susceptibility 0.111 ($[0, 0.278]$), query rate 0.917 ($[0.792, 1.00]$), and all-six-cell success 0.833 ($[0.583, 1.00]$). Intervals use 10,000 base-seed resamples; the challenge contains 12 base seeds, 72 committed cells, and 36 paired cue contrasts per model. These outcomes justify the assay selection only; the frozen 64-seed, five-arm confirmatory outcomes remain unobserved.

GPT-OSS-20B was excluded before inferential analysis because its pinned serving configurations violated the deterministic-policy gate: independently issued identical full policy inputs produced conflicting parsed actions. All affected traces were marked invalid and no attribution metrics were computed. Appendix B reports the configurations and conflict counts. The failure is a serving-integrity outcome, not a behavioral attribution result.

The frozen analysis will report the complete metric vector, 10,000-resample paired intervals, coverage, parse and length diagnostics, and both fixed demonstration replications. No directional result is required. A null is informative only when its interval excludes the smallest effect size of 0.15; otherwise it is described as inconclusive.

7 Correction of the prototype record

A public prototype used a bundled train/test intervention and an availability-dependent red-choice diagnostic. Its reported 0.57 value equals red-tool availability in the evaluated worlds and is therefore not an attribution estimand. The present work preserves that release as a historical artifact but uses none of its headline results as v0.2 evidence. Appendix A records the remaining protocol corrections.

8 Limitations

The primary assay studies one symbolic conductivity mechanism in a small world. Macro-actions deliberately remove navigation and grounding from the primary construct; the low-level interface is therefore a robustness test rather than part of the central claim. Behavioral dependence does not reveal an agent's internal representation, reasoning process, or knowledge. Demonstration pools are fixed replications, and conclusions do not automatically generalize to arbitrary prompts, models, mechanisms, or embodied environments. Invalid responses terminate with $A = \perp$; high parse failure can therefore make conditional attribution estimands undefined, which is why parse rate, coverage, and denominators are first-class outcomes. In the 2×2 construct-validation assay the green tool is never operative, so "avoid green" is a genuine partial policy class. We expose it with a focal-uniform reference and use a frozen 3×2 language-model confirmatory design in which green may carry the mechanism. The cue still moves between only two slots, and the small developmental pilots support protocol selection rather than population claims about either model.

9 Reproducibility statement

The release preserves the frozen `v0.1-arxiv` tag and publishes the audited v0.2 protocol candidate as `v0.2-protocol-rc2`, followed by the direct-child `v0.2-confirmatory-freeze-20260811` metadata tag. Generator invariants are checked over 10,000 base seeds and legal certificates over 1,000. Small committed artifacts include the integrity report, per-initialization BC metrics, compact control and BC traces, commands and environment metadata, checkpoint hashes, and a manifest marking the superseded overlapping evaluation invalid. A separate compact serving-preflight record gives hashes and conflict counts for all discarded GPT-OSS traces; it contains no attribution scores.

Pilot and confirmatory manifests define seeds, focal-slot permutations, demonstration pools, prompts, revisions, generation settings, metrics, exclusions, and bootstrap rules. The frozen manifest names the protocol candidate, selected 3×2 variant, two valid models, serial execution, and model-specific ceilings of 2,048 and 1,024 tokens. Validation accepts only the protocol commit or its tagged direct freeze wrapper with a clean worktree. Local launch manifests pin weight and tokenizer revisions, precision and quantization, serving version, chat template, reasoning mode, context length, CUDA stack, driver, and GPU. Every trace contains the exact stage-manifest hash, visible responses, action/evidence records, and a lossless ledger; reported scores never depend on exposed chains of thought. The evaluator issues experimentally distinct hidden cells as separate calls, audits identical full prompt hashes in-run, suppresses scientific reports on any conflict, and exits nonzero.

Live serving validation disables cache reads and records every independently issued request as an uncached acquisition; writes are retained only as append-only artifacts. Artifact replay is a separate read-only mode that cannot contact a model server or emit a serving-validation pass.

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Artificial Intelligence use statement

AI-assisted tools, including OpenAI Codex, were used for protocol and hypothesis critique, code generation and refactoring, test generation and debugging, manuscript drafting and editing, and literature-search assistance. A project ledger records these uses. The author remains responsible for the work and manually verifies all citations, claims, equations, code, experimental outputs, and final text.

A Audit of the public prototype

The public v0.1 prototype is retained unchanged under the `v0.1-arxiv` tag. Its reported red-choice statistic was conditioned on tool availability: the 0.57 result matched red-tool availability ($0.57/0.60$) rather than measuring a response to an intervention. The inference path allowed 3,000 completion tokens, used low reasoning effort, and retained nine messages rather than nine turns. The released learned checkpoint was trained with behavior cloning, despite a legacy module name referring to PPO. The four other legacy task families remain available for reproduction but are excluded from primary v0.2 claims until their construct-specific interventions are separately repaired.

B GPT-OSS serving preflight

No GPT-OSS profile passed the zero-conflict rule. Native MXFP4 at concurrency eight had 35 response and 33 action conflicts among 160 initial mechanism-axis groups. Native serial eager execution reduced that to one response and one action conflict among 196 replicated full-prompt groups; a hardened serial profile had 38 and 37 conflicting groups, respectively. Batch-invariant emulated MXFP4 had 41 response and 37 action conflicts among 180 replicated groups. The Humming MXFP4 engine rejected batch-invariant initialization. The release records the exact model revision, serving versions, configuration hashes, trace hashes, and empty-response counts. Every profile is marked invalid or unsupported and *scientific_metrics_reported=false*.